

## Lithium Iron Phosphate(LiFePO4 Battery)

# LFP 51.2V 5KWh-30KWh

### Features of LiFePO4 Battery

- **Longer Cycle Life:** Offers up to 20 times longer cycle life and five times longer float/calendar life than lead acid battery, helping to minimize replacement cost and reduce total cost of ownership.
- **Lighter Weight:** About 40% of the weight of a comparable lead acid battery. A drop in replacement for lead acid batteries.
- **Higher Power:** Delivers twice power of lead acid battery, even high discharge rate, while maintaining high energy capacity.
- **Wider Temperature Range:** -20°C~60°C
- **Superior Safety:** Lithium Iron Phosphate chemistry eliminates the risk of explosion or combustion due to high impact, overcharging or short circuit situation.

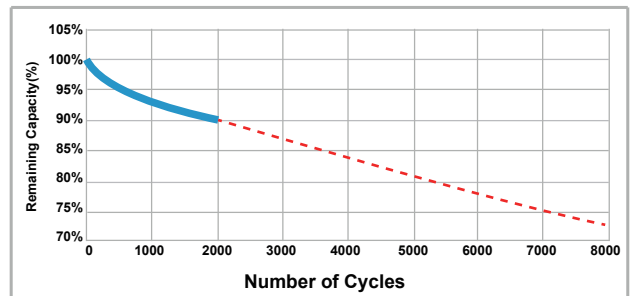


### Application

- Electric vehicles, electric mobility
- Solar/wind energy storage system
- UPS, backup power
- Telecommunication
- Medical equipment
- Lighting

### Specification

### Cycle Life Curve

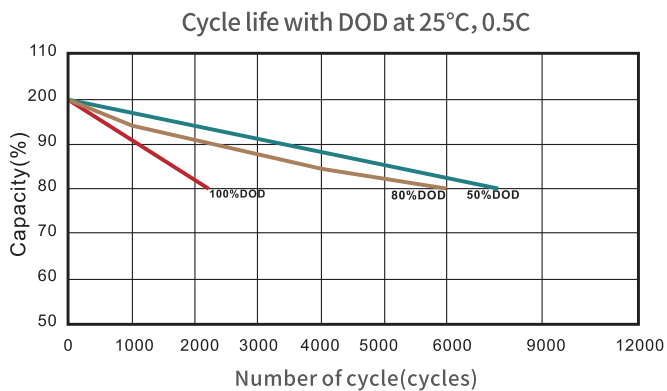


|                            | MODEL                     | RT-SLB-51.2V5kwh                                      | RT-SLB-51.2V10kwh | RT-SLB-51.2V15kwh | RT-SLB-51.2V20kwh | RT-SLB-51.2V25kwh | RT-SLB-51.2V30kwh |
|----------------------------|---------------------------|-------------------------------------------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| Electrical Characteristics | Nominal Voltage           | 51.2V                                                 |                   |                   |                   |                   |                   |
|                            | Nominal Capacity          | 200AH                                                 |                   |                   |                   |                   |                   |
|                            | Energy                    | 10240Wh                                               |                   |                   |                   |                   |                   |
|                            | Internal Resistance       | ≤500mΩ                                                |                   |                   |                   |                   |                   |
|                            | Cycle Life                | >6000 cycles @0.2C                                    |                   |                   |                   |                   |                   |
|                            | Months Self Discharge     | <3%                                                   |                   |                   |                   |                   |                   |
|                            | Efficiency of Charge      | 100% @0.2C                                            |                   |                   |                   |                   |                   |
|                            | Efficiency of Discharge   | 96-99% @1C                                            |                   |                   |                   |                   |                   |
| Standard Charge            | Charge Voltage            | 58.4±0.2V                                             |                   |                   |                   |                   |                   |
|                            | Charge Mode               | 0.2C to 58.4V, then 58.4, charge current 0.02C(CC/CV) |                   |                   |                   |                   |                   |
|                            | Charger Current           | 20A                                                   | 40A               | 60A               | 80A               | 100A              | 100A              |
|                            | Max. Charge Current       | 50A                                                   |                   |                   |                   |                   |                   |
|                            | Charge Cut-off Voltage    | 59.2V±0.2V                                            |                   |                   |                   |                   |                   |
| Standard Discharge         | Continuous Current        | 100A                                                  |                   |                   |                   |                   |                   |
|                            | Max. Pulse Current        | 150A(<3s)                                             |                   |                   |                   |                   |                   |
|                            | Discharge Cut-off Voltage | 43.2V                                                 |                   |                   |                   |                   |                   |
| Environmental              | Charge Temperature        | 0°C to 45°C(32F to 113F)@60±25% Relative Humidity     |                   |                   |                   |                   |                   |
|                            | Discharge Temperature     | -20°C to 60°C(-4F to 140F)@60±25% Relative Humidity   |                   |                   |                   |                   |                   |
|                            | Storage Temperature       | 0°C to 40°C(32F to 104F)@60±25% Relative Humidity     |                   |                   |                   |                   |                   |
|                            | Water Dust Resistance     | IP20                                                  |                   |                   |                   |                   |                   |
| Mechanical                 | Recommend Inverter        | 6.2kw                                                 | 6.2kw             | 8.2kw             | 8.2kw             | 10.2kw            | 10.2kw            |
|                            | Dimensions (in./mm.)      | 565*484*395                                           | 565*484*540       | 565*484*685       | 565*484*830       | 565*484*975       | 565*484*1120      |
|                            | Protocol (optional)       | RS485/RS232/CAN                                       |                   |                   |                   |                   |                   |
|                            | BMS                       | 16S100A                                               |                   |                   |                   |                   |                   |

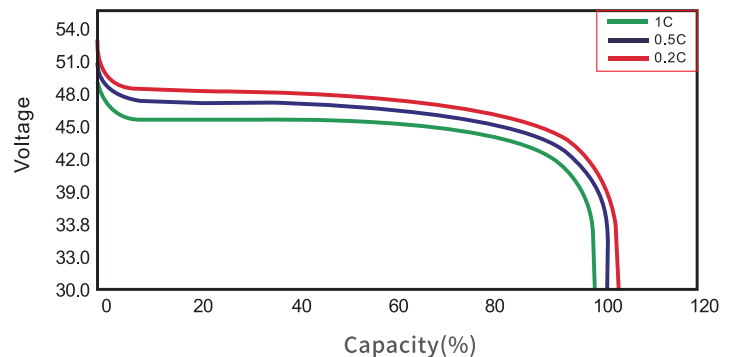
## LFP 51.2V 5KWh-30KWh

### Model Performance Diagrams

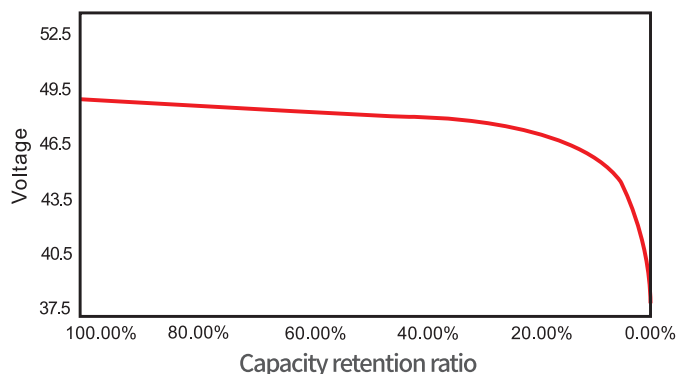
Number of Cycles Vs. DOD



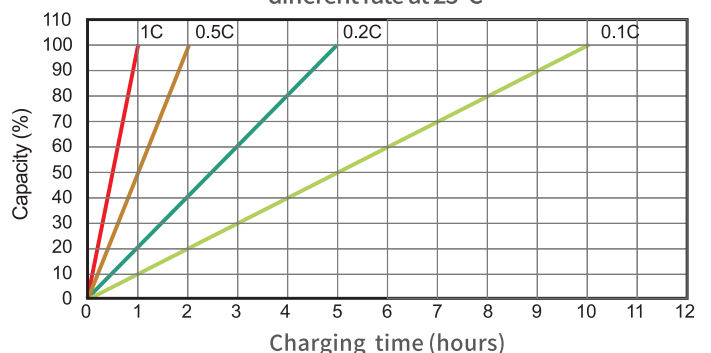
Discharge Performance at R.T.



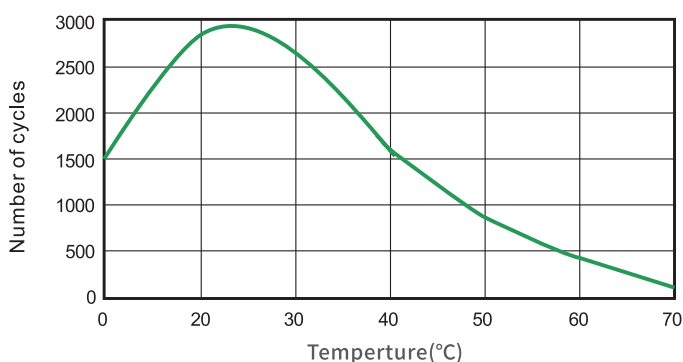
Battery Capacity (C) VS. Open Circuit Voltage (OCV)  
SOC Vs OCV



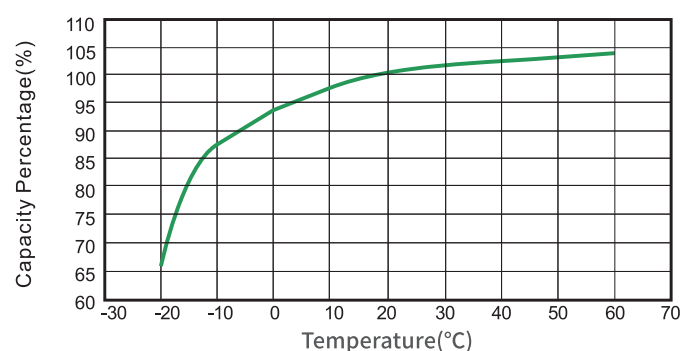
Battery Capacity Vs. Charging Time  
Charging capacity(%) VS time with different rate at 25°C



Cycle Life in Relation to Temperature



Temperature Effects on Capacity



**RESTAR SOLAR RENEWABLE ENERGY CO.,LTD**

Address: No.30 Yongshun Road, Zhulin Town, Jintan District, Changzhou, Jiangsu Province, China

Web: [www.restarsolar.com](http://www.restarsolar.com)

E-mail: [sales@restarsolar.com](mailto:sales@restarsolar.com)

Tel: +86 512-66292101

